



**Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)**

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member (s): Mrs.V.SrirenaNachiyar

Innovative Practice Description

- **Unit / Topic:** Unit I / Discrete Fourier Transform - Linear Filtering using FFT
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 4
- **Activity Chosen:** Concept Mapping
- **Justification:**

Collaborative learning is an educational approach to teaching and learning that involves groups of students working together to solve a problem, complete a task. Used to analyze FFT using linear filtering. Students must understand the concept of Linear Filtering.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**

First split the students into groups and ask them to sit as a group. The topic is divided into sub topics and each group of students allotted for sub topics. Students were asked to prepare the topic prior. They discussed within their group about the topic and present about the filtering techniques.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PSO1	PSO3
CO1	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9	PSO3
	2	1
Justification for correlation	In assignment the technical information will be interpreted in basic level and the same will be presented as a report. So, it is mapped at level 2	Students will design linear filtering to achieve the FFT algorithms. Hence level 1 is mapped.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students stated that they appreciated the activity and that it helped them remember topics and content when writing assessments.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the Linear filtering techniques using FFT. Due to this activity, the students were able to recall the topic and remember the important terms easily.

- ❖ ***Challenges faced in implementation:***

I have planned this activity for 40 minutes. Students take some time to forming the group. After formed the group, they discussed the topic. Students asked doubts regarding the topic before mapping the concept. I've given a brief overview of the FFT algorithms. I encouraged the students which it takes another 5 minutes for completing the activity.

References:

- ❖ Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.

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Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic:** Unit II / Infinite Impulse Response Filters – Characteristics of commonly used analog filters
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 5
- **Activity Chosen:** Think Pair Share
- **Justification:**

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. Activity is used to analyze the Characteristics of commonly used analog filters

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**

First split the students into groups and ask them to sit as a group. The topic is divided into sub topics and each group of students allotted for sub topics. Students were asked to prepare the topic prior. They discussed within their group about the topic and present about the filtering techniques.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO3
CO2	1	2	2	2	3	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Students discussed with their peers and present their views in front of all. So, it is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students stated that they felt the topic is very easy and understandable one & this activity helped them remember topics and content when writing assessments.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the characteristics of Analog filters. Due to this activity, the students were able to recall the topic and remember the important terms easily.

- ❖ **Challenges faced in implementation:**

Students took some time to forming the group. After formed the group, they discussed the topic. Students asked doubts regarding the topic before mapping the concept. I've given an overview of the Analog filters. I encouraged the students which it takes another 5 minutes for completing the activity.

References:

- ❖ Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.

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Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic:** Unit III / Finite Impulse Response Filters –Frequency Sampling Techniques
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Open Book Test
- **Justification:**

An open book exam is one in which students can give answers using their textbooks and notes. In the open book exam, memory and organizational skills are quite significant. This is because students must memorize each page of the textbook to determine which page of the textbook contains the solution to which question. This activity is used to understand the concepts and important terms in books

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

First split the students into groups and ask them to sit as a group. Students were asked to prepare the topic by using books in library. They discussed within their group about the topic and present about the frequency sampling techniques in paper.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO3
CO2	1	2	2	2	3	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Students discussed with their peers and write the topic in paper. So, it is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students stated that they felt that this activity helped them to refer various books for preparing notes about frequency sampling techniques. It is very useful to collect points from various books

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge of various frequency sampling techniques for processing the signal. Due to this activity, the students were able to recall the topic and remember the important terms easily.

- ❖ ***Challenges faced in implementation:***

Students took some time to forming the group in library. After formed the group, they discussed the topic using various Digital Signal Processing books in library. I encouraged the students for completing the activity with useful points.

References:

- ❖ Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.

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Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member (s): Mrs.V.SirengaNachiyar

Innovative Practice Description

- **Unit / Topic:** Unit IV / Finite Word Length Effects –Input /Output Quantization Errors
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen:** Jigsaw
- **Justification:**

Jigsaw is a cooperative group activity in which students are interdependent to achieve a common goal. In part one, each group is provided a different problems in Input /Output Quantization Errors. The group members become experts on that particular topic and create a group response.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

Students said that they felt this activity had improved their understanding of the steps and methodology for resolving problems in input/output quantization errors. During assessment tests, it is simple to recall the steps.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO3
CO2	1	2	2	2	3	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Students discussed with their peers and write the procedures and solving techniques in paper. So, it is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students stated that they felt that this activity helped them to understand the procedure and steps to solve the quantization errors. It is very useful to recollect the important steps for solving

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge about input/output quantization errors while processing the signal. Due to this activity, the students were able to recall the topic and remember the important terms easily.

- ❖ ***Challenges faced in implementation:***

Students took some time to forming the group. After formed the group, they discussed the topic and present the important terms in paper. I encouraged the students for completing the activity with useful points.

References:

- ❖ Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.

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Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member (s): Mrs.V.SrirenaNachiyar

Innovative Practice Description

- **Unit / Topic:** Unit V / Introduction to Digital Signal Processors –Application examples
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 21
- **Activity Chosen:** Flipped class
- **Justification:**

The flipped classroom model is based on the idea that traditional teaching is inverted in the sense that what is normally done in class is flipped or switched with that which is normally done by the students out of class. This Activity gives the clear understanding of the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

Students were given the task of preparing a presentation on the topic of "application examples of Digital Signal Processor". They described the architecture of various DSP processor used for several applications". Students were divided into groups and each group will prepared the detailed presentation of Application examples and deliver their presentation among all the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO3
CO5	2	2	3	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PO12
	2	1
Justification for correlation	Students will understand the concept, architecture of various DSP processor and will give clear information during discussion session. Hence it is mapped at level 2	Students were identify the previous instances in DSP processor where improvements in technology required lifelong learning for practitioners in accordance with changing. Thus

		level 1 is mapped
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The practise was well-received by students, who said it helped them remember subjects and substance while writing exams. They also noted that because they are presenting in front of all of the students, this activity boosts their confidence.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge of various DSP architecture. Due to this activity, the students were able to recall the topic and remember the important terms easily.

❖ **Challenges faced in implementation:**

I have planned this activity for 30 minutes. Students take some time to present the topic. After formed the group, they present the topic with presentation. Students asked doubts regarding the topic before presentation. I've given a brief overview of the Applications of DSP and encouraged the students to present which it takes another 5 minutes for completing the activity.

References:

- ❖ Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.

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